Conversion of Speech to Braille: Interaction device for Visual and Hearing Impaired

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***Abstract* – This research paper describes the implementation of a prototype device for both Visual and Hearing impaired persons get information from Speech converting to Braille output. For this interaction device HM2007 IC is used. The input is by receiving the speech signal and to recognize the digits. Once the spoken digit has been recognized, the DSP processer will send the corresponding ASCII code to the PIC with necessary encoded information. The output of this system is a Braille Display. The PIC microcontroller takes the encoded input from the DSP, decodes it and sends the appropriate recognized Braille cell trough the Solenoid Driver.**

The entire system is implemented with receiving digits of the Tamil Thirukural as speech signal and the output of the digits is tested with the Tamil Thirukural embossing every letter with delay time.

***IndexTerms* - Speech Recognition, PIC microcontroller, Braille, Solenoid drive**

I. INTRODUCTION

In recent years Speech recognition applications are becoming more and more essential. In real time practical sense applications, speech recognition used in solving issues, improving productivity which changes the style of human lives by human interaction system. Speech recognition considers many algorithms thatneed high computational requirements.

Speech Recognition system is one of natural communication that aimed to give an intelligent into computer to have an interaction with human. SR interacted with computer by using a voice through a microphone as an input device and resulting spoken word as an output converted by acoustic signal.

Speech Recognition system is a process that convertsthe speech signal into sequence of words.In order to develop techniques and systems for speech input to machine, by means of an algorithm implemented as a computer program.

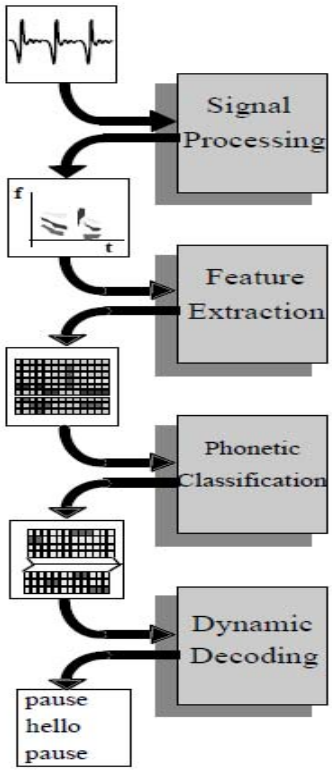

Fig. 1. Overview of the ASR process.

In future Speakerindependent voice recognition systems have a very strong probability it would be more convenient to use and would be able to improve productivity. Applying speech recognition for valuable applications identified two main

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factors that affecting the recognition rate in speech recognition, those are variability of acoustic which includes different pronunciation, accents, pitches, and volume; covers different speaking rates. Fig.1. Overviews the process of Automatic Speech recognition.

Braille script code

The Braille system comprises of a Braille cell per character consist of six embossed or raised dots arranged as shown Fig.2. The way the dots are raised gives meaning and each cell has the meaning of only separate letter, digit or simple punctuation mark.

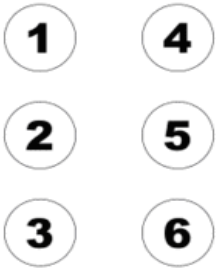

Fig. 2 Braille Cell Numbering

Braille codes are systems elegant, concise and very human systems for transcribing printed material using a Braille Alphabet. The Braille is a script used by visually impairedperson to read and write; the parameter name "dot-id" is used to define a braille cell. A Braille cell [1] is six dots arranged in 2 columns of 3 dots each. Each Braille cell as symbol represents a letter or a word or a combination of letters, symbols, numeral or punctuation mark.

The six dots are conventionally numbered from the left column as 1, 2, and 3 and from the right column, as 4, 5, and 6 as shown in Fig.2. The embossed or non-embossed of dots gives the Braille codeof the script. Today, in different languages different Braille codes are used to map character sets of to the six bit cells.

II. PROTOTYPE OVERVIEW

The prototype of this communicational device is a speakerindependent system that is used to receive connected spoken digits from 0 to 9 limited to four-digit number. On receiving the number, it performs the identification of respective Tamil Thirukural. Through solenoid drive the Tamil character is embossed as Tamil Braille script on Braille display at Real time. The block diagram of this real-time communicational device is shown below in Fig.3.

The idea of this device is to recognize the speech of one person and convert the data into Braille and embossed through Braille Display. This hardware would be in the assistance of Visual and hearing impaired people.

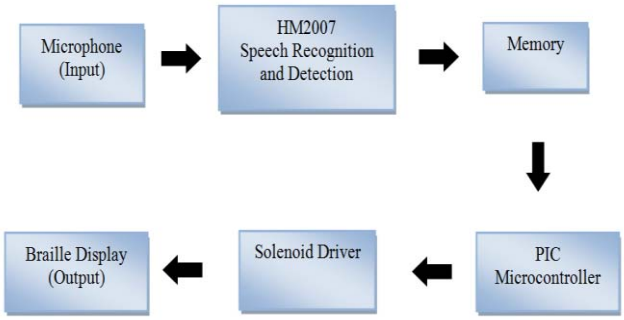

Fig. 3 Real Time Communication Device for Visual and Hearing Impaired Person

Specifications

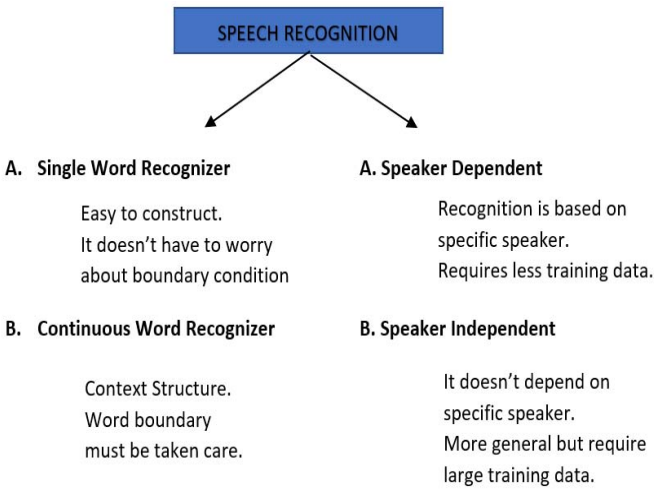

Fig.4 Classification of Speech Recognition

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Speech Recognition is categorized into two as

- Speaker dependent and
- Speaker independent.

Thissystem is a **Speaker independent** as to respond to regardless words of the speaker. Therefore, this system would respond to a large variety of speech. The word count of speaker independent is usually be lower than the speaker dependent, however within processing limits the high accuracy can still be maintain.

Recognition Style: Concerning the style of speech, Speech recognition systems have another constraint. The Recognition styles of speech are categorized: Isolated Speech, Connected Speech and Continuous Speech.

Connected Speech: This speech recognition system is optedas connected speech style as it is a point between isolated speech and continuous speech recognition. By connected speech the users can speak multiple words.

Speech Recognition Communication device:

Now a day, Speech recognition became the popular method of choice to control many electronic appliances like toys, tools and computers.The main part or heart of the system is the HM2007 IC. It can recognize more than 10words. TheSpeech recognition system implemented is a speaker-independent system, whereby the system is trained to recognize any individual voice.

Microcontroller & Solenoid driver

Microcontroller unitis the next important part of this system. The output of the chip goesto the decoder (micro-controller) [2] [3] which here acts as a Solenoid Drive unit. The output from the ports of HM2007 chip provides output as respective to the digits spoken by one person the interfacing unit would identifies the respective Tamil Thirukural spelled and convert the Tamil Character into Tamil Braille Script as shown in Fig.4.

Solenoid Driver and Braille Display:

The Braille Cell consists of 3X2 dots for representing the Braille projections of corresponding character. Each dot in the Braille cell consists of solenoid actuator which is coupled to the parallel output port of the Solenoid driver. There are totally 6 solenoid actuators, over them lies the dots. The solenoid actuator gets actuated once they get the 5V power supply through PIC to those 6 Solenoid (Braille Display) which corresponds the dots to be projected out, acts as the output device.

Tamil Braille Script

The aksharas of Indian languages are divided into vowels and consonants. Across the languages of Indian language, the number of vowels would be within 13 and 18 where the number of consonants would be within 33 and 37. Thus,nearly 50 Braille cells have been assigned for the basic vowels and consonants among 64 cells leaving the rest for special symbols.

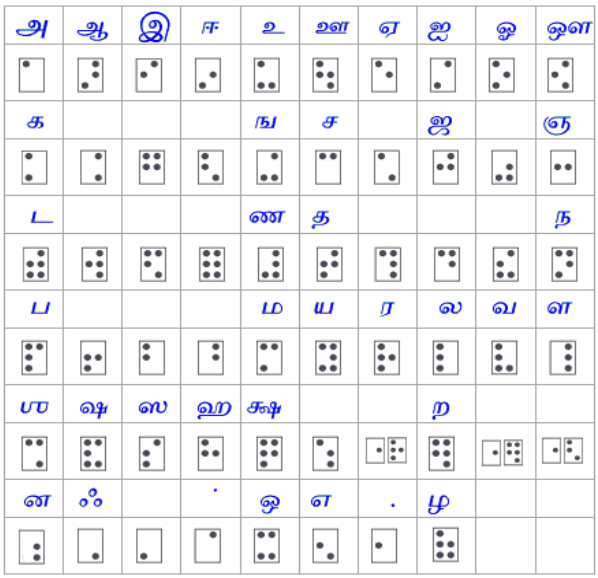

Fig5 Tamil Braille[1]

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Tamil language script consists of twelve vowels, eighteen consonants and 1 special character, the āytam. The product of both vowels and consonants results to 216 compound characters.Thus only 38 cells [18] have been assigned. The assignment of the cells is therefore applicable across all the languages though it must be stated that a few cells have to interpret based on the language.

III. IMPLEMENTATION

The implementation of Speech recognition system has following steps.

Step 1: The initial step is to capture the samples of raw speech waves as in Fig 6

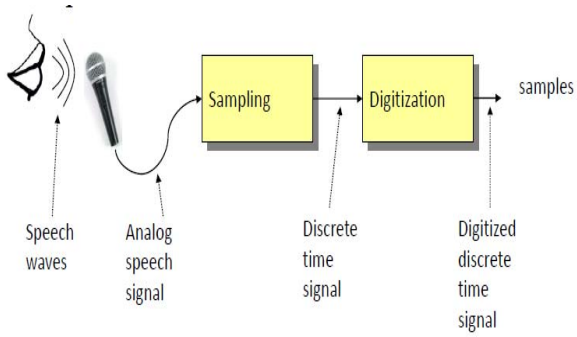

Fig6. Speech signal Capture

Step 2: End Point detection as it is in need to identify the presence of speech signal in a captured sampleas in Fig.7 by whichanalysis and recognizing process of speech in non-speech regions can be avoided.

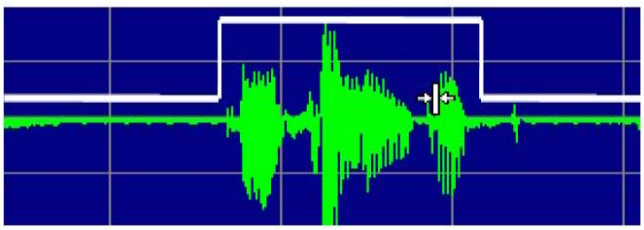

Fig7. Endpoint Identification

Step 3: Speech Recognition

There are three types ofspeech recognitionmodels:

–*Acoustic*type used formodellingspeech signals

–*Pronunciation*type models

–*Language*type used for modelling the structure of sentences

Acousticmodels, Pronunciation models and Language type models would be the inputs to the Speech Recognizer (Decoder) as in Fig 8.

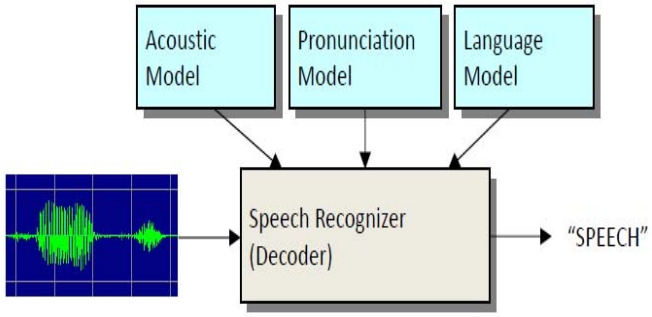

Fig8. Speech Recognizer

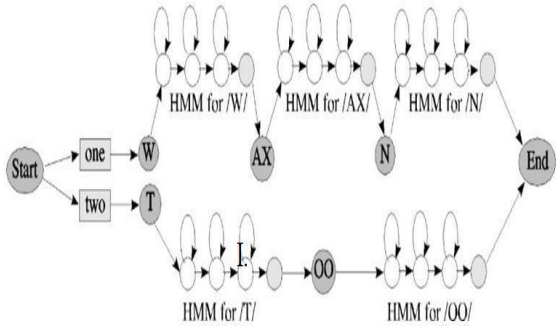

Fig.9 Example Search Graph

Step 4: HMM for Acoustic model

The *Search algorithm*is referred algorithm used by the recognizer as Fig. 9, Illustrate the SearchGraph used during the decoding process with sample of Tamil Thirukural with the number of digits limited to four digits as in Fig.10.

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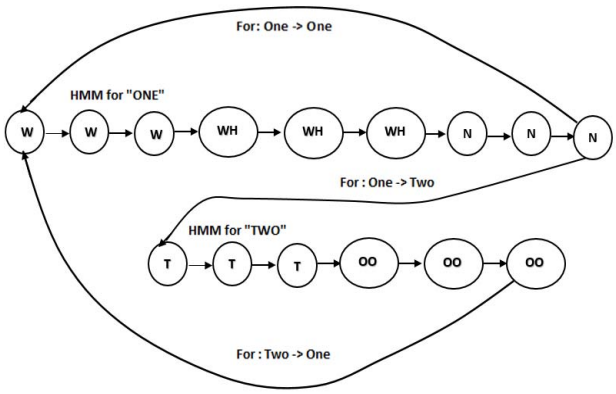

Fig. 10 Search graph for given vocabulary

The Search Graph is a directed graph composed of optionally emitting Search States and Search State Arcs with transition probabilities. Each and every state present in the graph will be able to represent components from the Language Model (words in rectangles), Dictionary (sub-word units in dark circles) or Acoustic Model (HMMs).

Step 5: Feature Extraction: The pre-recorded samples of the words are considered as training data [8][9].

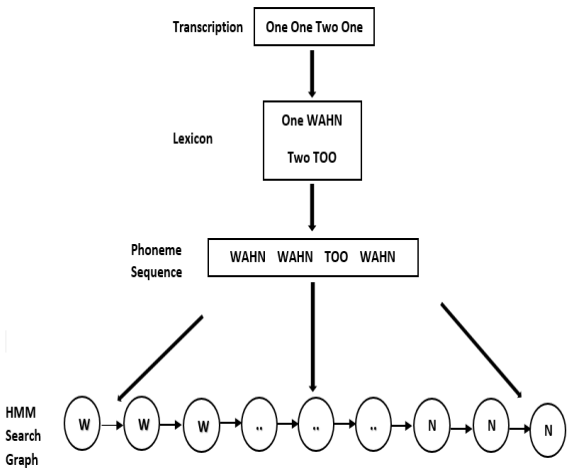

Fig.11 Map word from vocabulary to HMM

Step 6: Result of the HMM: The single recognition unit or *hypothesis* would be result of Speech Recognition systems.

Step 7: Post Processing:The decoded [12] result be processed under Tamil Thirukuraldatabase and identify the Tamil Thirukuralrespectively.

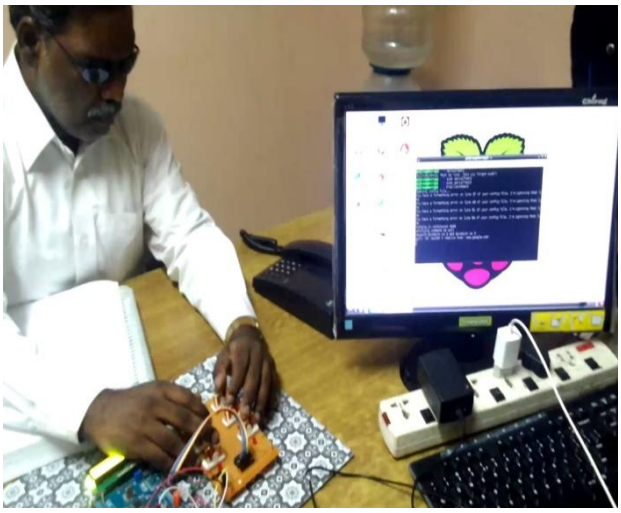

Fig. 12 Snap shot of Real Time Experiment

Step 8: Solenoid drive: Through solenoid drive the Tamil character is embossed as Tamil Braille script on Braille display[5][6][15].

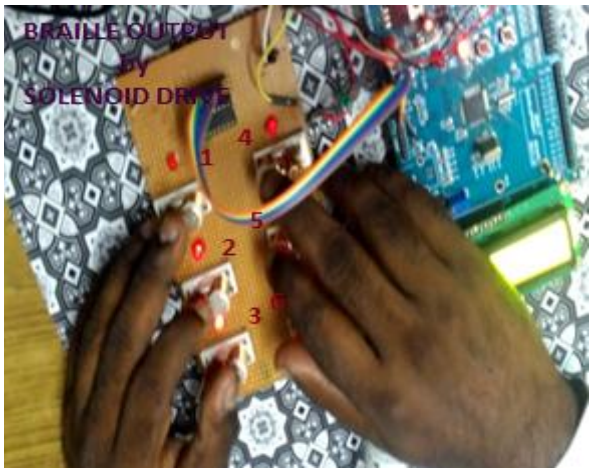

Fig.13 Snap shot of Real Time Experiment

An assistive device for a disabled person can be assessed by the way of handling and level ofpriorities analyzed by low error rate, and high efficiency. This device is implemented using a Single Board Computer Raspberry Pi, as to be portable and usage of Real time experiment.

Main reason of using the device[10][13]would be decided on analyzing the advantage of the device should be higher than the disadvantage encountered. After implementation of the

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prototype, it is tested with few visually impaired and both visual and hearing impaired persons.

IV. CONCLUSION

The main purposeof this research paper was to implement and demonstrates the use of a communication device used for and by the visual and hearing impaired persons.

Due to dual sensory deprivation, visual and hearing impaired person face many difficulties on interaction with normal person in the world, without an assistantthey can't communicate with others.

The benefits of this prototype allow them to interactwith others without any assistant. Inspite of that the above methodology has a setback, before testing the people expecting intimation and information of Tamil Braille script included.

V. REFERENCES

[1] BRL: Braille through Remote Learning - http://brl.org/refdesk/brlcont.html

[2] Cater, John P., Electronically Hearing: Computer Voice and speech Recognition,Howard W Sams& Co., Indianapolis, IN, 1984.

[3] Fourcin, A., G. Harland, W. Barry, and V. Hazan, editors, Voice and speech Input and Output Assessment, Ellis Horwood Limited, Chichester, UK, 1989.

[4] VandanaPushe and Rupinderdeep Kaur, "Issues in Converting Punjabi to Braille", Asian Journal of Computer Science and Information.

[5] VandanaPushe, Nidhi Bhalla and Rupinderdeep Kaur, "Architecture of Gurmukhi to Braille conversion system", International Journal of Computer Science and Inormation Technology & Security(IJCSITS), Vol.2, No.2, April 2012, pp. 467-771

[6] Wu Xiaosong, Zhu Haihong, Kim Seong-Hyok, Allen Mark G., "A Portable Pneumatically-Actueted Refreshable Braille Cell", The 14th International Conference on Solid-State Sensors, Actuators and Microsystems, Lyon, France, June 10-14, 2007.

[7] C´elineMancas-Thillou, Silvio Ferreira, Jonathan Demeyer, Christophe Minetti, and Bernard Gosselin "A Multifunctional Reading Assistant for the Visually Impaired" in Hindawi Publishing Corporation EURASIP Journal on Image and Video Processing Volume 2007, Article ID 64295, 11 pages doi:10.1155/2007/6429.

[8] Fezari, M.; Khati, A.E, "New Speech Processor and Ultrasonic sensors based embedded system to improve the control of a motorized wheelchair," Design and Test Workshop, 2008. IDT 2008. 3rd International 20-22 Dec. 2008; pg.:345- 349

[9] M. A. M. Abushariah, T.S.Gunawan, and O.O.Khalifa, "English Digits Speech Recognition System Based on Hidden Markov Models," Proceedings of International Conference Computer and Communication Engineering (ICCCE), 2010, pg. 1-5.

[10] Saad Mad S., M. Hussein, M. S. YaacobA. R. Musa,F. Razaly, M. Z. MD Zain, , M. Y. Abdullah, "Development of Piezoelectric Braille Cell Control System Using Microcontroller Unit (MCU)", Issue 6, Volume 9, June 2010

[11] J.Bhattacharya, S.Majumder, G.Sanyal, "Automatic inspection of Braille Character: A vision based approach." International Journal of computer & Organization trends – Volume 1 Issue 3 – 2011.

[12] Blenkhorn, Paul; "System for Converting Print into Braille", IEEE Transactions on Rehabilitation Engineering, Volume.5, No. 2, 1997, pg. 121-129.

[13] H. Tehranchi, S. Sunthankar, R. malyan," Reading aid for the Blind", Image processing for disabled people, IEEE Colloquium on 2 oct 1992.

[14] Bliss James C, "A Relatively High Resolution Reading Aid for the Blind", IEEE Transactions on Man-Machine Systems, Volume MMS-10, No. 1, March 1996

[15] Iain Murray, and Andrew Pasquale (2006), "A Portable Device for the Translation of Braille to Literary Text", Proceedings of ACM Conference on Assistive Technologies (ASSETS), pp. 231- 232

[16] Koo1Igmo, Jung Kwangmok, Koo Jachoon, Jea-do Nam, Lee Youngkwan and Choi HyoukRyeol, "Wearable Fingertip Tactile Display", SICE-ICASE International Joint Conference 2006 in Bexco, Busan, Korea.Oct. 18-21, 2006.

[17] Rozeha A. Rashid, Nur HijaMahalin, MohdAdibSarijari and Ahmad Aizuddin Abdul Aziz, "Security System Using Biometric System: Design and Implementation of Voice Recognition System (VRS)", Proceedings of the International Conference on Computer and Communication Engineering, pp 898-902, ISBN :978-1-4244-1692-9, May 2008.

[18] Capt.Dr.S Santhosh Baboo, V.Ajantha Devi; "COMPUTER BASED TAMIL BRAILLE SYSTEM - A Review";11th International Tamil Internet Conference, Annamalai University, Chidambaram, India , Jointly conducted by International Forum for Information Technology (INFITT) and Annamalai University Dec 28-30, 2012. Pg.57 -61

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